# Xiongwei Luo

AI Product & Application Development

Zurich, Switzerland | +86 137 8924 2010 | xiongweiluo1@gmail.com
Internship / Working Student | GitHub | Portfolio

## EDUCATION

**ETH Zurich** | **MSc Integrated Building Systems** Sep 2025 - Jul 2027 (expected)
Coursework: Machine Learning, Hands-On Deep Learning, Scientific Machine Learning, Building Agentic AI Systems.

**Tongji University** | **Bachelor of Architecture** Sep 2019 - Jul 2025
Urban data analysis and human-computer interaction; academic scholarships in four consecutive years.

**National University of Singapore** | **Young Fellowship** Jul 2025
Summer programme on interdisciplinary AI.

## TECHNICAL SKILLS

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|---|---|
| **Programming** | Python, SQL |
| **AI and agents** | MCP/FastMCP, Function Calling, JSON Schema, QLoRA, Harness evals, paired evaluation |
| **Frameworks and tools** | React, FastAPI, Node.js, Docker, Git, GitHub Actions, Figma |
| **Data and systems** | MySQL, SQLite, Redis, Celery, WebSocket, automated testing |

## PROFESSIONAL EXPERIENCE

**JD.com** | **Product Design and Prototyping Intern** Jun 2026 - Sep 2026
Autonomous Vehicle Product Team

- Built a unified B2B prototype with **17 top-level menus and 91 navigation entries**; integrated 54 entries with existing prototypes and documented requirements for the remaining 37.
- Implemented deep links, history navigation and unsaved-form protection; established **91-page screenshot audits** and interaction checks, and delivered an offline HTML export.

## SELECTED PROJECTS

**RhinoCoder** | **Verifiable AI Agent for 3D Modelling**
Independent project | GitHub

- Designed a planning, execution and verification loop with **23 versioned MCP tools**, scene inspection and geometric assertions; added cancellation, undo, rollback, idempotency and privacy checks.
- Passed **270/270 real-Rhino runs** on a fixed benchmark of 30 tasks, three configurations and three repetitions; broader generalisation remains unverified.
- QLoRA improved exact tool-call sequences from **20/80 to 77/80** on offline tasks. The real-Rhino quality gate remains unmet; the default production route is unchanged.

**LabelHub** | **AI Annotation and Quality Platform**
Team project | Schema architecture, core development & integration | Mock demo

- Built shared schema contracts and a designer-compiler-renderer pipeline; applied preflight checks to field suggestions and Function Calling to AI pre-review, with reviewer diffs for oversight.
- Implemented asynchronous pre-review and exports with retries, idempotency and status tracking; delivered Docker Compose deployment and CI with **nine real-backend E2E scenarios passing**.

**Family Garden** | **Collaborative AI Game**
Team project | AI, backend & game systems | Live demo

- Built AI memory and image-analysis workflows with schema validation and moderation; implemented member identity, family data isolation, private photo storage and WebSocket presence.
- Used transactions and operation identifiers for inventory, planting and harvesting; validated the Web release with **26/26 regression checks, 4/4 production E2E scenarios** and iPhone Safari checks.

## LANGUAGES

Chinese (native) | English (IELTS 7.0) | German (B1)